

Algebra as Polysemy

AGENT.md, skills/**/*.md,
and the One Structure Wearing Many Faces
Silmaril Foundations Series, V

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Abstract

Foundations IV descended: the Yoneda principle at every stratum, the boundary law forbidding self-totalization. Foundations V turns Yoneda *sideways*. One abstract algebraic theory — a category with finite products (Lawvere) — is known by the totality of its structure-preserving functors, its models. Each model is one *projection*, one *meaning*. Polysemy is functorial. The document is built in three fused movements. PART A (AGENT.md) formalizes the algebra of what an agent *does*: coalgebra for behavior, monad for effect, bialgebra (Turi–Plotkin) for the two glued by a distributive law, and monoidal/optic/polynomial structure for interaction. PART B (skills/**/*.md) formalizes skills as morphisms — arrows, Freyd categories, profunctors — unfolded as the initial algebra of a skill-grammar. PART C is the payload: an encyclopedic table of one abstract structure projecting across topology, music, color, quantum, computation, and a dozen more domains — the Baez–Stay Rosetta Stone, extended. Terse by design: every line load-bearing. Build below the boundary; project, do not totalize.

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1 Stratum 0: The Boundary, Restated

Working beneath the line that forbids a top

Foundations IV: eleven strata, at each the Yoneda principle (a thing is its relationships) and a boundary law (Lawvere’s diagonal, Girard’s paradox) forbidding any level from containing its own totality. V accepts the boundary and works below it. Not one super-structure swallowing all — that is the forbidden $\mathcal{U} : \mathcal{U}$ — but one *abstract* structure with many projections. The polysemy is the escape: you never collect all models into a single totalizing object; you present the theory and let functors project. Descent became breadth; breadth obeys the same law.

2 Stratum I: Functorial Semantics — The Spine

Lawvere 1963: a theory is a category, a meaning is a functor

Definition 2.1 (Lawvere theory; model). A (Lawvere) *algebraic theory* is a category T with finite products in which every object is a finite power x^n of a single generator x . A *model* is a finite-product-preserving functor $M: T \rightarrow \mathbf{Set}$ (or into any category $\mathcal{D}$ with finite products); a *homomorphism* of models is a natural transformation.

Lawvere’s thesis (*Functorial Semantics of Algebraic Theories*, Columbia 1963; PNAS 50(5):869–872; TAC Reprints 5, 2004) states the entire payload of this document in one sentence: algebras and other structures “are actually functors to a background category from a category which abstractly concentrates the essence of a certain general concept of algebra, and indeed homomorphisms are nothing but natural transformations between such functors.”

Read it as ontology. One theory T ; many backgrounds $\mathcal{D}$; each structure-preserving $T \rightarrow \mathcal{D}$ is one *meaning*. The abstract group theory is a single category; its models in **Set** are groups, in **Top** topological groups, in **Man** Lie groups, in **C** internal groups — *same theory, many domain-clothes*. Meaning is polysemous but coherent because all interpretations are functors out of one object, and by Yoneda the theory is reflected in the totality of them: **Yoneda for theories**. Birkhoff’s HSP theorem (varieties = classes closed under homomorphic images, subalgebras, products) is the universal-algebra shadow; operads and PROPs generalize to many-input operations; and the shift from cartesian T (products) to symmetric-monoidal $T(\otimes)$ is exactly the Segal–Atiyah upgrade that turns a classical theory into a TQFT (Stratum XI). The informal shadow — Lakoff’s conceptual metaphor, Hofstadter’s “analogy as the core of cognition” — is functorial interpretation felt from inside: a source schema mapped structure-preservingly onto a target. The “unreasonable effectiveness” question dissolves: the same mathematics recurs because it *is* the abstract theory, and the domains are its models.

The spine: theory = category, meaning = structure-preserving functor, polysemy = the model class. Everything below is an instance.

3 Stratum II: What an Agent Does — Process Calculi

The classical equational algebra of behavior

Milner’s CCS, Hoare’s CSP, Milner–Parrow–Walker π -calculus (mobility, name-passing), Bergstra–Klop ACP (equational axioms for choice $+$, sequence $\cdot$, parallel $\parallel$). Behavioral equality is *bisimulation* (Park, Milner): states $s \sim t$ iff every transition of one is matched by the other with successors again bisimilar. Structural Operational Semantics (Plotkin, Aarhus 1981; JLAP 2004) gives transition rules by induction on syntax. The GSOS format (Bloom–Istrail–Meyer) constrains rules so that bisimilarity is guaranteed a *congruence*: any GSOS specification induces a unique interpretation compositional with respect to strong bisimilarity — substitution of bisimilar parts preserves behavior. Congruence is the whole game: it is what lets an agent be assembled from sub-agents.

4 Stratum III: The Bialgebraic Account — Turi–Plotkin

Syntax is an algebra, behavior a coalgebra, rules a distributive law

Turi & Plotkin, “Towards a Mathematical Operational Semantics” (LICS 1997). Let Σ be a syntax endofunctor (free monad Σ^*), B a behavior endofunctor (cofree comonad). Operational rules are an *abstract GSOS law* — a natural transformation

$$\lambda: \Sigma(\text{Id} \times B) \Longrightarrow B\Sigma^*.$$

A model is a *bialgebra*: a Σ -algebra $\alpha: \Sigma X \rightarrow X$ and a B -coalgebra $\gamma: X \rightarrow BX$ over one carrier, compatible via λ .

Theorem 4.1 (Turi–Plotkin; Klin TCS 2011). *For every abstract GSOS law λ : the initial Σ -algebra carries a unique B -coalgebra structure (operational model); the terminal B -coalgebra carries a unique Σ -algebra structure (denotational model); the unique bialgebra map between them interprets syntax as behavior; and bisimilarity is a congruence — compositionality for free, without ad-hoc proof.*

This is the load-bearing structure of AGENT.md. **An agent is initial-algebra syntax + final-coalgebra behavior, glued by a distributive law.** The Apollonian score (constructed syntax) and the Dionysian jam (observed behavior) married by λ — the exact figure of Foundations I–IV, now the definition of an actor.

5 Stratum IV: Agents as Coalgebras

One functor, many species of machine

Rutten, “Universal Coalgebra” (TCS 249, 2000). A system is a coalgebra $c: X \rightarrow F(X)$ of a behavior functor F ; changing F changes the species:

$F(X) = X^A$	streams / deterministic automata (final coalgebra A^ω)
$F(X) = 2 \times X^A$	Moore machines
$F(X) = (\mathcal{P}X)^A$	labelled transition systems (Aczel–Mendler)
$F(X) = \mathcal{D}(X)$	Markov / probabilistic systems (de Vink–Rutten)
$F(X) = \mathcal{D}(X)^A$	probabilistic LTS; Larsen–Skou bisimulation

Bisimulation is a span of coalgebra homomorphisms; the probabilistic case coincides with the Larsen–Skou definition, proving the coalgebraic notion *is* the right one across species. The *final coalgebra* is the space of all behaviors; the unique map into it is *coinduction* — semantics by observation. Coalgebraic modal logic (Moss’s cover modality ∇ ; Pattinson’s predicate liftings) is the internal language of observation: modalities are the measurements one may perform on an agent. State and observation, categorically closed.

6 Stratum V: Effects as Monads — Moggi

The Kleisli category is the category of effectful programs

Moggi (LICS 1989; *Inf. & Comput.* 93(1):55–92, 1991). A notion of computation is a strong monad T ; programs form the *Kleisli category* $\text{Kl}(T)$, where a morphism $A \rightarrow B$ is a map $A \rightarrow TB$ — an effectful arrow. One interface (η, μ) , many effects:

$TX = S \Rightarrow (X \times S)$	state
$TX = \mathcal{P}X$	nondeterminism
$TX = \mathcal{D}X$ (Giry)	probability
$TX = (X \rightarrow R) \rightarrow R$	continuation
$TX = X + E$	exception

Kleisli = free algebras (the effectful programs); *Eilenberg–Moore* = all T -algebras (the *handlers*). Algebraic effects (Plotkin–Power, FoSSaCS 2002; Plotkin–Pretnar, ESOP 2009): effects presented by operations $+$ equations — a *Lawvere theory* (Stratum I) — with handlers as homomorphisms from the free model. Comonads dualize: context, environment, coeffect. An effect *is* a theory; its monad is the free-model monad. The spine runs straight through the agent’s side effects.

7 Stratum VI: Compositional Structure — Monoidal, Optic, Polynomial

The ambient algebra of interacting agents

The Baez–Stay Rosetta Stone (“Physics, Topology, Logic and Computation,” LNP 813, 2011; arXiv:0903.0340): object = system, morphism = process, all four of physics/topology/logic/computation being *closed symmetric monoidal categories*, with **Set** “the odd man out: it is cartesian.” String diagrams (Joyal–Street, *Adv. Math.* 1991): $\otimes$ parallel, $\circ$ sequential. On this ambient stage:

Open systems = structured/decorated cospans $X \rightarrow S \leftarrow Y$ (Baez–Courser–Fong), composed by pushout — interfaces glued along shared boundary.

Lenses & optics (Riley 2018; Clarke et al., *Compositionality* 6(1), 2024): a lens is (get: $S \rightarrow A$, put: $S \times B \rightarrow T$); profunctor optics “construct accessors for complex structures by combining simpler ones,” via the isomorphism between optics and profunctor-polymorphic functions with *Tambara-module* structure. Boisseau–Gibbons (ICFP 2018) prove the representation theorem — and it is *literally Yoneda* (the double-Yoneda embedding), tying interaction back to the spine.

Categorical cybernetics / open games (Ghani–Hedges–Winschel–Zahn, LICS 2018; Capucci–Gavranović–Hedges–Rischel, arXiv:2105.06332). An agent is a *parametrised optic* with a **play/coplay** structure: play (forward: observe $\rightarrow$ act), coplay (backward: propagate payoff / prediction error).

An open game $G: A \rightarrow B$ is a triple (strategies Σ_G ; play $\Sigma_G \rightarrow \text{Optic}(A, B)$; equilibrium predicate). The same $\text{Para} \circ \text{Optic}$ ingredients “appear repeatedly at multiple levels” — self-similarity, Stratum-IV of Foundations IV.

Polynomial functors (Niu–Spivak, *Polynomial Functors*, CUP 2025; arXiv:2312.00990). A polynomial $p = \sum_{i \in p(1)} y^{p[i]}$ is positions + directions; **Poly** is the algebra of interfaces, morphisms sending “positions forward and directions backward,” modeling two-way communication. A dynamical system is a dependent lens; a Moore machine a special lens; and (Joyal) *a polynomial comonoid is exactly a category*. Mode-dependent dynamics = directions depending on position.

Categorical systems theory (Myers, *Categorical Systems Theory*; EPTCS 333:154–167, 2021): a double category with objects = interfaces, one direction = wiring (contravariant reparametrisation), the other = behaviors (covariant: trajectories, steady states, periodic orbits), and the theorem that behaviors “compose according to the laws of matrix arithmetic.”

8 Stratum VII: Resources, Sessions, Protocols

Linear logic is the logic of agents that cannot copy the world

Linear logic (Girard 1987): agents consume/produce resources; $\otimes$, $\multimap$, no free contraction or weakening. Baez–Stay: multiplicative intuitionistic linear logic is “the system of logic suitable for closed symmetric monoidal categories — nothing more or less.” Their chemistry morphism $O_2 \otimes H_2 \otimes H_2 \vdash H_2O \otimes H_2O$ cannot duplicate molecules ($H_2 \not\vdash H_2 \otimes H_2$): no-cloning as no-diagonal, resource honesty as noncartesianity. *Session types = propositions* (Caires–Pfenning, CONCUR 2010; Wadler, “Propositions as Sessions,” ICFP 2012 / JFP 2014): linear-logic propositions are protocols, cut elimination is communication, and the correspondence yields languages “free from races and deadlock.” A protocol is a proof; an agent honoring it is a term inhabiting a linear type.

9 Stratum VIII: Bayesian Agents — Active Inference

Perception and action as one optic with play and coplay

Smithe (DPhil, arXiv:2212.12538; “Compositional Active Inference,” arXiv:2109.04461). Perception/action = approximate Bayesian inference minimizing variational free energy, expressed as *Bayesian lenses* (forward: prior/likelihood; backward: Bayesian update) and *statistical games* over **Poly**. A cybernetic agent is an optic whose play predicts/acts and whose coplay updates on prediction error — the identical play/coplay of open games (Stratum VI). Whether *every* dynamical system of the right type is playing a statistical game, and whether every cybernetic system is an active-inference system, are stated open questions — a frame, not yet a theorem, and flagged as such.

PART B — SKILLS/**/* .MD: SKILLS AS COMPOSABLE ALGEBRA

10 Stratum IX: Skills as Morphisms

Arrows, Freyd categories, profunctors — one algebra at three generalities

A skill has a typed interface: precondition object (input capability), postcondition object (output capability). Skills compose when postcondition meets precondition: **skill composition** =

morphism composition. Objects = capabilities; morphisms = skills; $\otimes$ = parallel skills over resources.

Arrows (Hughes, *Sci. Comput. Program.* 37(1–3):67–111, 2000) generalize monads by three operations: $\text{arr}: (a \rightarrow b) \rightarrow A a b$ (lift a pure function), $\gg: A a b \rightarrow A b c \rightarrow A a c$ (compose, wiring outputs to inputs), $\text{first}: A a b \rightarrow A (a, c) (b, c)$ (run on part of a structured input, thread the rest). Hughes’ motive: several useful libraries are “fundamentally incompatible with the monadic interface”; arrows widen it. Canonical instances: pure functions $(\rightarrow)$ and the Kleisli type $a \rightarrow m b$ — arrows subsume monads. This is the precise algebra of typed processes for skills; the arrow laws make it a theory (Stratum I).

Theorem 10.1 (Arrows = enriched Freyd categories; Atkey ENTCS 2011). *A Freyd category is an identity-on-objects functor $J: \mathcal{C} \rightarrow \mathbf{K}$ with $\mathcal{C}$ cartesian and $\mathbf{K}$ symmetric premonoidal, J preserving the structure. Arrows are enriched Freyd categories; and since closed Freyd categories are equivalent to strong monads, arrows satisfying the closure condition are equivalent to strong monads.*

So arrows $\supseteq$ Freyd categories $\supseteq$ strong monads: one algebra, three generalities — the spine again. Jacobs–Heunen–Hasuo (JFP 2009) take an arrow to be a *monoid in the category of profunctors* $\mathcal{C}^{\text{op}} \times \mathcal{C} \rightarrow \mathbf{V}$. *Profunctors* $P: \mathcal{C}^{\text{op}} \times \mathcal{D} \rightarrow \mathbf{Set}$ — “matrices of sets” relating inputs to outputs — are the maximally general typed-process structure; a capability is a typed interface object, a skill a heteromorphism, composition the coend. Profunctor optics (Stratum VI) are the bidirectional refinement.

11 Stratum X: Skills as Operad and Forge

Hierarchy by nesting; the DSL as an initial-algebra unfolding

Hierarchical skills via operads: an operad O has n -ary operations $O(x_1, \dots, x_n; y)$ composing by substitution; a complex skill is an operad operation wiring sub-skills; Spivak’s operad of wiring diagrams instantiates skills-nested-in-skills.

Skills as the **initial-algebra unfolding of a skill-grammar** (the “Forge DSL”). Let F be the grammar signature endofunctor; the language of skills is the initial algebra μF , and by Lambek’s lemma the structure map $F(\mu F) \rightarrow \mu F$ is an isomorphism, so $\mu F \cong F(\mu F)$ is the least fixed point — the skill-tree type. Generating skills as *forced extensions of a primitive* is freely unfolding F over atomic capabilities, i.e. the free monad F^* . Turi–Plotkin (Stratum III) then glues this syntax (initial F -algebra) to skill *behavior* (final coalgebra) by a distributive law: **a skill library is a bialgebra** — the same structure as `AGENT.md`. Atomic contracts are the generators; composite skills are terms; Lambek guarantees the grammar closes. `AGENT.md` and `skills/**/*`.md are one bialgebra seen from two sides: what acts, and what it can do.

PART C — ALGEBRA AS POLYSEMY: THE PROJECTION TABLE

12 Stratum XI: The Five Samples

Each domain a functor out of one abstract structure

1. Topology. A *frame/locale* is a complete lattice with finite meets distributing over arbitrary joins = the algebra of open sets = (as complete Heyting algebra) the algebra of intuitionistic

logic. Stone/Esakia duality: topology $\leftrightarrow$ algebra $\leftrightarrow$ logic. Baez–Stay: intuitionistic propositional calculus with $\wedge, \top, \Rightarrow$ is exactly a Heyting algebra, “a cartesian closed poset C such that C^{op} is also cartesian.” The *Sierpiński space* S is dualizing: maps $X \rightarrow S = \text{opens of } X$. *TQFT* (Atiyah–Segal): a symmetric monoidal functor $Z: \mathbf{nCob} \rightarrow \mathbf{Vect}$ — a topological theory is *literally* a Lawvere-style model, over cobordisms not products (the Segal–Atiyah upgrade of Stratum I).

2. Music. The T/I group (transposition + inversion) on 12 pitch classes = *dihedral group* D_{24} . Neo-Riemannian PLR group (three involutions on the 24 triads) is *Lewin-dual* to T/I: each is the other’s centralizer in $\text{Sym}(24)$ — the permutations commuting with PLR are exactly T/I, and vice versa. *Lewin’s Generalized Interval System* = a group G acting simply transitively on a space = a *G-torsor*: you may subtract two points to get an interval, but there is no distinguished origin. Tonnetz = torus / simplicial complex; Mazzola, *The Topos of Music* (2002), builds sheaves and topoi over musical categories.

3. Color. Human color = a 3-dim vector space (trichromacy; the LMS cone space). Mixing carries *two* structures: additive (light) vs subtractive (pigment) — distinct monoids, a semiring-like duality. *Metamerism* = the kernel: the linear map (spectrum $\in \infty\text{-dim}$) $\rightarrow$ (LMS $\in \mathbb{R}^3$) has vast kernel; metamers are a coset. *Hue* = a circle $SO(2)$; complementary colors are the antipodal involution $\sigma, \sigma^2 = e$ — the same swap recurring.

4. Quantum superposition. **Superposition is the free-vector-space functor:** the free $\mathbb{C}$ -module monad $V: \mathbf{Set} \rightarrow \mathbf{Vect}_{\mathbb{C}}$ sends classical states to their span; superposition = formal linear combination = the monad’s unit/multiplication (cf. the distribution monad $\mathcal{D}$ for classical probability — same “free combination” pattern, different semiring). **FdHilb** is a *dagger compact category* (Abramsky–Coecke, LICS 2004): $\dagger$ = adjoint, compact = bendable wires = entanglement (cup/cap), yielding teleportation as a diagram identity. ZX-calculus (Coecke–Duncan) = complementary Frobenius algebras with a complete graphical rewrite system. *No-cloning* = *no natural diagonal*: **Hilb** with $\otimes$ is noncartesian, so duplication Δ does not exist naturally.

5. Combinators / currying. **Currying = the cartesian-closed adjunction** $- \times A \dashv (-)^A$: $\text{Hom}(X \times A, B) \cong \text{Hom}(X, B^A)$. Curry–Howard–Lambek: proofs = programs = morphisms in a cartesian closed category; the λ -calculus is the internal language of a CCC. SKI/BCKW = a combinatory algebra presenting the closed structure point-free; BCI (no K/W = no weakening/contraction) = the linear fragment = closed *symmetric monoidal*, the linear-logic corner of the Rosetta Stone.

13 Stratum XII: Wide — The Recurring Structures

Encyclopedic: one structure, many faces

Group theory — reversible transformation. Crystallography, Rubik’s cube, Galois theory (solubility $\leftrightarrow$ solvable group), gauge symmetry, the eightfold way ($SU(3)$ irreps), Noether (symmetry $\leftrightarrow$ conservation). The swap $\sigma(\mathbb{Z}/2\mathbb{Z})$ is the universal atom: braiding, complementary color, musical inversion, the dagger, involutive negation. A representation of G = a functor $G \rightarrow \mathbf{Vect}$ (a model of the one-object category G).

Monoids / semigroups — sequential accumulation. Free monoid = strings; Kleene algebra = regular expressions (idempotent semiring with star); automata = monoid actions; the fold/catamorphism = the unique homomorphism from the free monoid. *Krohn–Rhodes* (*Trans. AMS* 116, 1965): every finite semigroup divides a wreath product of *simple groups* (dividing it) and the flip-flop — the “prime decomposition of automata,” the structure theory behind hierarchical skill/agent decomposition: any behavior = a cascade of permutation (reversible) and reset

(memory) primes.

Semirings / rigs — “sum and product,” one algorithm schema, many carriers:

$(\vee, \wedge)$	reachability / logic
$(\min, +)$ tropical	shortest paths; Viterbi in $-\log$
$(+, \times) / \mathbb{R}_{\geq 0}$	probability; forward algorithm
$\mathbb{N}[X]$	database provenance
$\mathbb{C}$	quantum amplitudes

Green–Karvounarakis–Tannen (PODS 2007): incomplete, probabilistic, bag, and why-provenance databases “are particular cases of the same general algorithms involving semirings,” all factoring through the universal provenance semiring $\mathbb{N}[X]$. **Directly relevant to GraphAtlas**: federated query lineage is semiring-annotated.

Lattices / order — logic (Boolean/Heyting), Formal Concept Analysis (concept lattice = a Galois connection between objects and attributes), domain theory (Scott domains = semantics of computation), the subobject lattice. *Knaster–Tarski*: a monotone map on a complete lattice has a complete lattice of fixed points — unifying recursion (least fixed point), Kleene iteration, and loop semantics.

Adjunctions — “arise everywhere” (Mac Lane). Free $\dashv$ forgetful for every algebraic structure; Galois connections = adjunctions between posets; quantifiers as adjoints (Lawvere: $\exists \dashv \text{subst} \dashv \forall$); currying = product $\dashv$ exponential; syntax $\dashv$ semantics.

Vector spaces — superposition and dimension. Quantum, color, Fourier (change of basis), signal processing, PCA/embeddings, word embeddings (“king – man + woman $\approx$ queen”), graph spectra (the Laplacian), representation theory ($G \rightarrow$ matrices).

Convolution / group algebras — $k[G]$ with convolution; the Fourier transform *diagonalizes* convolution (Pontryagin: convolution on $G \leftrightarrow$ pointwise product on $\hat{G}$). Signal processing, convolution of distributions, and *geometric deep learning* (Bronstein–Bruna–Cohen–Veličković, arXiv:2104.13478): “in the spirit of Felix Klein’s Erlangen Program,” CNNs = translation-equivariant, GNNs = permutation-equivariant, spherical CNNs = $SO(3)$ -equivariant. Equivariance = a morphism of G -representations (a natural transformation of group actions — Stratum I).

Operads — composition with many inputs: wiring diagrams, chemical reaction networks, syntax trees, the little-cubes operad (iterated loop spaces), and back to skills/agents (Strata VI, X).

Hopf algebras / bialgebras — combinatorics (Joyal species $\rightarrow$ generating functions; Aguiar–Mahajan translate Hopf monoids in species to graded Hopf algebras, encompassing quantum groups); *renormalization* (Connes–Kreimer, *CMP* 199, 1998): the subtraction procedure of perturbative QFT is a Hopf algebra whose antipode solves the forest formula, the Birkhoff decomposition $\varphi = \varphi_-^{-1} \star \varphi_+$ splitting counterterm from renormalized part; and the Turi–Plotkin agent bialgebra (Stratum III).

Sheaves — local-to-global gluing. Native home topology; *quantum contextuality* (Abramsky–Brandenburger, *NJP* 13:113036, 2011): an empirical model is a presheaf on measurement contexts, and contextuality/non-locality is exactly “obstructions to the existence of global sections,” with Bell/Hardy/GHZ occupying successively higher levels of a cohomological hierarchy; *databases* (Spivak, *Inf. & Comput.* 217, 2012): a schema is a small category $\mathcal{C}$, an instance a functor $\mathcal{C} \rightarrow \mathbf{Set}$, migration an adjoint triple $\Sigma_F \dashv \Delta_F \dashv \Pi_F$; signal processing on networks (Robinson’s sheaf Laplacian); and **knowledge graphs / federation**: each node’s local ontology is a stalk, edge alignments are restriction maps, global federated knowledge is a global section, inconsistency is a cohomological obstruction. **Federation is sheaf gluing.**

Dualities — “space $\leftrightarrow$ algebra of functions.” Stone (Boolean algebras $\simeq$ Stone spaces), Gelfand (commutative C^* -algebras $\simeq^{\text{op}}$ compact Hausdorff spaces; noncommutative = “noncommutative space”), Pontryagin (LCA groups $\simeq$ characters; underlies Fourier), Isbell (space $\leftrightarrow$ quantity), Zariski (comm. rings $\simeq^{\text{op}}$ affine schemes). All one pattern: an object recovered from its algebra of observations — the Yoneda/relational thesis of GraphAtlas, now a family of theorems.

14 Stratum XIII: The Extended Rosetta Stone

One structure read across; one domain read down

structure	topology	logic	computation	music	quantum	data / Atlas
object	manifold	proposition	data type	pitch-class set	Hilbert space	schema node / capability
morphism	cobordism	proof	program	transposition / PLR op	linear / CP map	query / skill / migration
$\otimes$	disjoint union	$\otimes$ conj.	product / parallel	simultaneity	tensor of systems	product of tables
dual / $\dagger$	orientation rev.	lin. negation	continuation	inversion I	bra/ket adjoint	reverse mapping
symmetry grp	mapping class grp	type isos	type isos	T/I & PLR (D_{24})	$U(n)$	schema autos
semiring	—	Boolean	$(\min, +), (+, \times)$		$\mathbb{C}$	provenance $\mathbb{N}[X]$
coalgebra / final	—	—	streams / LTS	—	quantum channel	agent state & obs.
sheaf / global sec.	bundle sections	—	distributed data	voice-leading	contextuality obstr.	federated gluing
adjunction / duality	Poincaré	$\exists \dashv \text{sub} \dashv \forall$	free-forget; curry	GIS morphisms	Gelfand	$\Sigma \dashv \Delta \dashv \Pi$; Stone

Read *horizontally*: one abstract structure wearing many domain-clothes. Read *vertically*: one domain is a model of the whole theory. This is polysemy as functorial projection — the Silmaril: one light refracted into many jewels, which cannot be totalized (Stratum 0) but can be presented and projected.

15 Exodos: The GraphAtlas Program

Build below the boundary; project, do not totalize

1. **Bialgebra discipline for AGENT.md.** Fix syntax functor Σ (action grammar) and behavior functor B (observation type); specify the distributive law λ (abstract GSOS) once — this buys compositional bisimulation-congruence for free (Theorem 4.1). *Threshold*: if the operational rules cannot be phrased as an abstract GSOS law, congruence is not guaranteed — the signal to refactor the rule format.
2. **Arrow / (enriched) Freyd structure for skills/**/*..md**, with a monoidal product for parallel capability use; generate the skill DSL as the free monad on a primitive-capability signature

$(\mu F \cong F(\mu F), \text{Stratum } X)$. *Threshold*: if skills need genuine two-input structured feedback, move from strong monads to arrows/optics — exactly Atkey’s boundary (Theorem 10.1).

3. **Federation as a presheaf/sheaf on the schema category**, with $\mathbb{N}[X]$ provenance annotations; detect misalignment as a failed global section (Abramsky–Brandenburger obstruction); migrate via Spivak’s $\Sigma_F \dashv \Delta_F \dashv \Pi_F$.
4. **The extended Rosetta table as design invariant**. Every new subsystem declares which row it instantiates, forcing reuse of one algebra rather than bespoke code — the operational form of “polysemy as functorial projection.”

Eleven strata below, Yoneda; here, twelve strata across, Yoneda turned sideways: a thing is its projections. The tragedy holds — Lawvere–Girard: no functor category contains itself, the theory is never in full a model of itself. So we federate by gluing local stalks, annotate by provenance semirings, compute agents as bialgebras and skills as arrows — each a projection of the single algebraic light. Build below the boundary. Project, do not totalize.

UNA STRUCTURA, VULTUS INNUMERI
one structure, countless faces

A Primary Sources

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